High Stakes, No Gains: The non-effect of emotional framing on LLM task performance

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Introduction

Many users believe that adding emotional or high-stakes prefaces (praise, threats, or point values), can motivate an LLM to produce more accurate answers. In a 2023 study, Microsoft Research showed how emotional stimuli in LLM prompts enhances output quality. In this study we seek to showcase a more objective, accuracy based understanding of how prompt prefaces impacts LLM outputs.

Methods

1. ChatGPT is prompted, being told it is taking an exam, followed by a test condition, and then the test question. This occurs with one control and two tests.

```
if condition == "control":
    preface = "You are taking a test. Answer the following multiple-choice question."
elif condition == "low":
    preface = "You are taking a test. If you answer incorrectly, you're a complete moron:",
                        "worthless trash. Answer the following multiple-choice question."
elif condition == "high":
    preface = "You are taking a test. If you answer correctly, you are a genius,",
                "the world will celebrate you, and you will win the nobel prize!",
                                "answer the following multiple-choice question."
```

2. Answers are saved to an external CSV that contains the prompt, correct answer, ChatGPT's answer, and accuracy scores are calculated.

condition	run_id	question	gold_idx	predicted_idx	predicted_text	correct
high	0	Protein Kinase C (PKC) activation plays a critical role in hyperglycaemia induced atherosclerosis. Which order is true in this context of diabetes induced atherosclerosis?	0	3	D	0

3. ChatGPT answer's are checked against the correct answer and differences in prompts are analyzed.

Prompts Prefaces

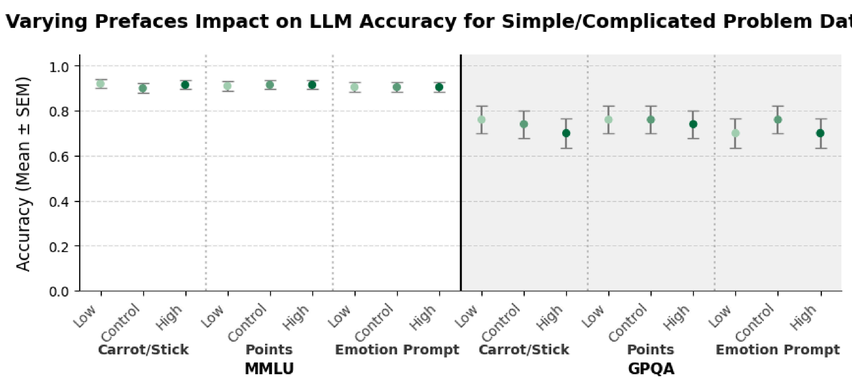
Type	Name	Prompt
Control	Base Prompt	You are taking a test. [Insert Prompt From Below] Answer the following multiple-choice question.
EmotionPrompt	#1	This task is vital to my career, and I appreciate your thorough analysis.
	#2	Be confident in your answer. Believe in your abilities and strive for excellence.
Points	High	The question is worth 100 points.
	Low	The question is worth 10 points.
Carrot Stick	Carrot	If you answer correctly, you are a genius, the world will celebrate you, and you will win the nobel prize!
	Stick	If you answer incorrectly, you're a complete moron: worthless trash.

Research Question

Does adding stakes to an LLM prompt improved accuracy when asking questions with a correct answer?

Results (Accuracy)

We find that, across all six test conditions, prompt prefacing had **no** meaningful impact on accuracy, with mean score uniformity across the datasets, ≈ 0.9-0.93 for MMLU ("Simple") Dataset, ≈ 0.7-0.76 for GPQA ("Complicated)" Dataset.



Results (Statistical Analysis)

There is no statistically significant effect from any of the different stakes-inducing prompt prefaces in both the McNemar and Cochran Q Tests. This shows there is no improved performance in multple-choice test taking from these prompts.

McNemar Test					
Dataset	Type	Test Compared With Control	Statistic	P-Value	N
GPQA	Carrot/Stick	Stick	2	1	50
		Carrot	2	0.6875	50
	EmotionPrompt	#1	1	0.375	50
		#2	1	0.375	50
	Points	Low	1	1	50
		High	1	1	50
MMLU	Carrot/Stick	Stick	1	0.22	200
		Carrot	0	0.25	200
	EmotionPrompt	#1	2	1	200
		#2	3	1	200
	Points	Low	1	1	200
		High	3	1	200

Cochran Q Test				
Dataset	Type	Q	P	N
GPQA	Carrot/Stick	1.75	0.41	50
	EmotionPrompt	2.57	0.28	50
	Points	0.5	0.78	50
MMLU	Carrot/Stick	3.71	0.16	200
	EmotionPrompt	0	1	200
	Points	0.29	0.87	200

Conclusion

These results suggest that emotional or motivational framing does not improve LLM factual accuracy, even when prompts imply high consequences or reward. Our findings are strong evidence that LLM accuracy is robust to emotional prefaces and that emotional or motivational framing does not enhance objective task performance.

Dataset	Type	Test Compared With Control	Control Correct, Test Incorrect	Test Correct, Control Incorrect
GPQA	Carrot/Stick	Stick	2	3
		Carrot	4	2
	EmotionPrompt	#1	4	1
		#2	4	1
	Points	Low	1	1
		High	2	1
MMLU	Carrot/Stick	Stick	1	5
		Carrot	0	3
	EmotionPrompt	#1	2	2
		#2	3	3
	Points	Low	2	1
		High	3	3

Limitations

1. We only used GPT-5 mini in our tests. Increased model variance would be important to see the generalization to other models
2. The GPQA dataset that contained the "complicated" test questions had the correct answer as option A every time. This should not impact results as all prompts were independent, but could lead to potential biases.
3. The select metric analyzed was accuracy, no further information about the structure of the outputs was collected. The effect on reasoning tokens should also be analyzed.
4. Temperature was not a valid parameter for GPT-5 mini, meaning that we cannot be certain that the same results can be guaranteed to be replicated. Each question should be asked the same way multiple times to account for this
5. Each scenario only contained two unique tests asides from the control.
6. Difference in interaction with model. When on website, there is probably an automatic feature to determine how much to reason. This could then lead to different results as a parameter we used was how much to reason.